

ScholAR: Page-Grounded Retrieval-Augmented Generation for Scientific Document Comprehension

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Abstract

Understanding and querying scientific papers demands precise, evidence-grounded comprehension that general-purpose language models struggle to provide reliably. Existing PDF-chat systems hallucinate citations, discard document structure, and cannot trace generated claims back to verifiable source pages. We present **ScholAR** (Smart Companion for **H**olistic **O**rganization, **L**iterature **A**nalysis & **R**esearch), a retrieval-augmented generation framework designed specifically for scientific document comprehension that runs **entirely on-device**, with no cloud API in the loop for either text or vision inference. ScholAR enforces strict page-boundary-preserving text segmentation, ensuring that every retrieved chunk maps exactly to a single PDF page. An indirect citation grounding mechanism decouples LLM generation from page-number prediction, replacing fragile in-context page references with verifiable evidence identifiers that the frontend resolves to precise PDF highlights. Beyond single-document text QA, ScholAR extends to **visual grounding** (routing figure/table questions to a local, natively multimodal model) and **multi-document citation-chasing** (resolving and ingesting a paper’s own references so questions can be answered against both the anchor paper and its cited works). We introduce a manually annotated retrieval benchmark spanning 14 query–chunk pairs across 3 landmark NLP papers, evaluated using Recall@ K and Mean Reciprocal Rank (MRR). Our BM25-primary retrieval system achieves Recall@5 of 0.929 and MRR of 0.788, outperforming keyword-overlap baselines by 10.1 percentage points in MRR. On an 18-case visual-grounding pilot benchmark spanning four reasoning types, retrieval correctly routes to the target figure or table in 18/18 cases. On a 10-cluster multi-document locality benchmark, ScholAR retrieves the correct cited paper within the top 5 results 80% of the time (MRR 0.356), though top-1 precision (10%) remains a clear open challenge, consistent with prior findings that even large foundation models struggle at cross-paper localization.

1 Introduction

- We propose **ScholAR**, an end-to-end, **fully local** RAG framework for scientific document comprehension that

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enforces strict page-boundary-preserving text segmentation and performs both text and vision inference on-device.

- We introduce an **indirect citation grounding** mechanism that decouples LLM generation from page number prediction, reducing hallucinated citations.
- We extend single-document grounding to **visual question answering** over figures and tables and to **multi-document citation-chasing**, resolving and ingesting a paper’s own references so cross-paper questions can be answered.
- We present quantitative benchmarks for all four capabilities (retrieval, faithfulness, visual grounding, multi-document locality), each with manually annotated cases and Recall@ K /MRR-style metrics.
- We conduct controlled **ablation experiments** comparing keyword overlap, BM25-only, and BM25-primary-with-reranking retrieval architectures, and compare two local models (qwen3.5:9b, gemma4:12b) on visual grounding to characterize model choice for a fully on-device system.

2 Related Work

2.1 Retrieval-Augmented Generation

The standard RAG framework was introduced by Lewis et al. (Lewis et al. 2020). BM25 probabilistic retrieval was formalised by Robertson and Zaragoza (Robertson and Zaragoza 2009). The BEIR benchmark (Thakur et al. 2021) provides a heterogeneous zero-shot evaluation suite for retrieval systems.

2.2 Scientific Literature RAG Systems

PaperQA (Lála et al. 2023) was an early high-accuracy RAG agent for scientific QA and literature synthesis, but relies on proprietary cloud LLMs for both retrieval agency and generation. OpenScholar (Asai et al. 2024) substantially outperforms PaperQA and GPT-4o on literature-synthesis quality using an open retrieval pipeline, though generation still runs through large hosted or self-hosted models rather than a single fully local system. SciRAG (Ding et al. 2025) further improves over both OpenScholar and PaperQA

via citation-graph-aware symbolic reasoning and outline-guided synthesis. None of these systems targets a fully on-device deployment: all either call a cloud API or assume access to large (multi-tens-of-billions-of-parameters) self-hosted models. ScholAR instead targets consumer hardware, running both text and vision inference through a single sub-15B-parameter local model, trading some raw capability for zero data exposure and zero API dependency.

2.3 Scientific Document Understanding

GROBID (Lopez 2009) and Nougat (Blecher et al. 2023) provide structured parsing of academic PDFs into machine-readable formats. SlideTailor (Zeng et al. 2026) generates personalised slides from scientific papers but relies on raw PDF text extraction, the same bottleneck ScholAR addresses. The S2ORC corpus (Lo et al. 2020) provides a large-scale resource of parsed academic papers.

2.4 Visual Document Retrieval

ColPali (Faysse et al. 2024) and its successors index each document page as a visual embedding, enabling page retrieval directly from image content without OCR or layout-aware text extraction. This is an architecturally different bet from ScholAR’s approach: rather than embedding whole pages as images, ScholAR keeps BM25 text retrieval as the primary signal and routes only figure/table-targeted queries to a vision-capable model applied to a cropped region around the identified figure. This keeps the common text-QA path fast and fully lexical while still enabling grounded visual question answering when a query names or implies a specific figure or table.

2.5 Multi-Modal and Multi-Document Scientific QA

M3SciQA (Li et al. 2024) is the closest existing benchmark to ScholAR’s multi-document extension: it pairs each anchor paper with its cited references and requires both multi-modal and cross-document reasoning, evaluating 18 foundation models and finding that all of them substantially underperform human experts at this task. ScholAR’s multi-document locality benchmark (Section 5.4) is smaller in scope but directly comparable in spirit, and our results echo M3SciQA’s central finding: cross-paper localization is hard even when the correct reference is already available in the retrieval pool.

2.6 Question Answering over Documents

QASPER (Dasigi et al. 2021) is the closest existing benchmark to our task, covering information-seeking QA over scientific papers. SciFact (Wadden et al. 2020) evaluates fact verification over biomedical literature.

2.7 Citation Grounding and Faithfulness

Gao et al. (Gao et al. 2023) study attribution in LLM generation; their analysis shows that citation quality degrades sharply without retrieval grounding. Min et al. (Min et al. 2023) propose FActScore, which decomposes generated text

Figure 1: ScholAR document processing and retrieval pipeline. The system extracts page-level text, applies page-preserving chunking, and serves grounded answers with indirect citation IDs.

into atomic claims and verifies each claim against a knowledge source, adopting this decomposition strategy in our benchmark construction. Laban et al. (Laban et al. 2022) introduce SummaC for consistency detection in summarization, with two variants: SummaC-Conv (fine-tuned NLI) and SummaC-ZS (zero-shot via sentence embeddings). We implement SummaC-ZS because it does not require a fine-tuned NLI model (e.g., DeBERTa-MNLI), making it reproducible without GPU infrastructure or API access. Zha et al. (Zha et al. 2023) show that embedding-based alignment metrics substantially outperform ROUGE-family metrics for hallucination detection; this motivates our choice of SummaC-ZS over lexical alternatives.

3 Problem Formulation

$$\hat{a} = \arg \max_a P(a \mid q, \mathcal{C}_q) \quad \text{where} \quad \mathcal{C}_q = \text{Retrieve}(q, \mathcal{C}, k) \quad (1)$$

4 Method

4.1 Document Processing Pipeline

Given a paper (an arXiv identifier or an uploaded PDF), ScholAR downloads or accepts the file, extracts per-page text via PyMuPDF, applies page-preserving chunking (below) and figure/table extraction (Section 4.5), and persists the results as `metadata.json`, `pages.json`, `chunks.json`, and `figures.json`. Figure 1 summarises the flow from raw PDF to a grounded, citable answer.

4.2 Page-Preserving Chunking

Standard character- or token-window chunking routinely splits a chunk across a page boundary, breaking the one-to-one mapping between a retrieved chunk and the PDF page a citation should highlight. ScholAR instead chunks strictly within each page: Algorithm 1 slides a window of $W=1400$ target words with $\delta=120$ -word overlap over each page’s token stream independently, so a chunk never spans two pages. A lightweight regex pass additionally tags each chunk with a detected section title (Abstract, Method, Results, ...) when a heading-like line is found near the top of the page; the retriever uses this as a section-hint signal below.

4.3 BM25-Primary Retrieval with Heuristic Reranking

The production retriever ranks chunks with Okapi BM25 (Robertson and Zaragoza 2009) ($k_1=1.4$, $b=0.72$) as the primary signal, then applies a small set of additive heuristic boosts: a query-expansion dictionary maps research-specific terms to related vocabulary (e.g., *result* → *accuracy*, *BLEU*, *score*); an explicit page number in the query

Algorithm 1 Page-Preserving Chunking

Require: Pages $P = \{p_1, \dots, p_n\}$, target words W , overlap δ

Ensure: Chunks $\mathcal{C}$

```
1:  $\mathcal{C} \leftarrow []$ 
2: for each page  $p_i \in P$  do
3:    $w \leftarrow \text{split}(p_i.\text{text})$ 
4:    $s \leftarrow 0$ 
5:   while  $s < |w|$  do
6:      $e \leftarrow \min(s + W, |w|)$ 
7:      $\mathcal{C}.\text{append}(\text{Chunk}(w[s:e], p_i.\text{page}))$ 
8:      $s \leftarrow e - \delta$ 
9:   end while
10: end for
11: return  $\mathcal{C}$ 
```

adds a page-hint boost; query language implying a paper section (e.g., *limitation*, *results*) adds a section-hint boost when a candidate chunk’s detected section matches; and rhetorical phrase cues (*we propose*, *we introduce*, *outperforms*) add a small boost, since claim-bearing sentences are more likely to satisfy an information-seeking query. Visual-cue boosting for figure/table chunks (Section 4.5) uses the same additive mechanism. We initially explored a design in which a dense/semantic score dominated ranking; our own retrieval ablation (Table 5) showed BM25-primary was the more reliable choice on this benchmark, so semantic, section, and phrase signals are kept as light rerankers rather than a primary signal, while dense retrieval is used only in the separate Hybrid BM25+Dense+RRF configuration evaluated for faithfulness (Section 5.2).

4.4 Indirect Citation Grounding

Free-form citation prompting — asking the model to write, e.g., “[p. 7]” directly in its answer — is fragile: the model has no reliable way to know which page a claim actually came from and will readily hallucinate a plausible-looking page number. ScholAR instead assigns each piece of retrieved evidence a short-lived evidence identifier ($E_1, E_2, \dots$) for the current turn, and the generation prompt instructs the model to cite *only* these evidence IDs, never a page number directly. After generation, ScholAR discards any page number the model wrote on its own (a direct violation of the instruction) and rewrites each valid evidence-ID citation into a compact numbered reference ($[1], [2], \dots$) in citation order; the frontend resolves each numbered reference back to its source chunk’s page and highlights the corresponding region of the rendered PDF. This decouples the model’s citation behavior from page-number prediction entirely: the model only has to decide *which* retrieved evidence supports a claim, while the mapping from evidence ID to actual page and highlight region is deterministic application logic rather than a model output.

4.5 Visual Grounding

When a query is judged to reference visual content (via cue terms such as “figure,” “table,” or “plot,” or an explicit fig-

ure/table number), retrieval boosts figure and table chunks so they can compete against much longer text chunks in the BM25 ranking. Each figure and table caption on a page is detected via regex, and the region above the caption is rendered at $3\times$ zoom and cached as a standalone image alongside the caption text. A query naming an explicit figure or table number (e.g., “explain figure 2”) is matched directly against that figure’s own label rather than relying solely on lexical overlap: this matters because our tokenizer, like most BM25 implementations, only begins a token on a letter, so “Figure 2” and “Figure 14” both reduce to the token `figure` once digits are dropped, making them otherwise indistinguishable to the ranker. When the top-ranked chunk is a figure or table, ScholAR base64-encodes the cached image and sends it, together with the caption and supporting text context, to the same local Ollama model used for text generation (qwen3.5:9b and gemma4:12b are both natively multimodal), rather than to a separate vision-specific model or a cloud vision API. If the image is missing or the model call fails, the system degrades to a caption-only extractive answer rather than failing outright.

4.6 Multi-Document Extension

ScholAR resolves an anchor paper’s own bibliography and lets the user ingest individual cited works into the same study session. For arXiv papers, the reference list is resolved via the Semantic Scholar API by arXiv ID; for uploaded PDFs without a known arXiv identifier, references are resolved via an S2 title search over the last several pages of the document, then regex-matched against both numbered ($[N]$) and author-year (Vaswani et al., 2017) in-text citation styles. Ingesting a reference downloads its PDF (when available), applies the same page-preserving chunking pipeline as the anchor paper, and tags every resulting chunk with a `source_paper_id`. At query time, chunks from all currently-loaded secondary papers are merged with the anchor paper’s chunks into a single retrieval pool, and each returned citation is labeled with its source paper so answers can distinguish anchor-paper evidence from evidence drawn from a cited work.

4.7 Study Goal Generation

ScholAR generates eight paper-specific study goals per document using a local Ollama model (qwen3.5:9b or gemma4:12b). Each goal has a title, a 2–4 sentence description grounded in paper text, and 3–5 recursive subquestions whose evidence chunks are resolved by a second BM25 retrieval pass. When no LLM is available, a fully extractive fallback generates goals directly from sentence patterns in the abstract and introduction.

4.8 Faithfulness Evaluation: SummaC-ZS + Semantic Coverage

We evaluate citation faithfulness using the SummaC-ZS protocol (Laban et al. 2022), the zero-shot variant of SummaC that does not require a fine-tuned NLI model. The key advantage of SummaC-ZS over ROUGE-family metrics is that it operates at the *semantic* level via dense embeddings, not

at the lexical level, making it robust to paraphrased hallucinations that ROUGE cannot detect.

Tier 1: SummaC-ZS (Semantic Entailment). Each ground-truth claim c is first decomposed into atomic sub-claims $\{a_1, \dots, a_m\}$ via sentence- and clause-level splitting (FactScore-style (Min et al. 2023)). For each atom a_i and each retrieved chunk $\mathcal{C}_j$, we compute:

$$s(a_i, \mathcal{C}_j) = \max_{t \in \text{sents}(\mathcal{C}_j)} \cos(\phi(a_i), \phi(t)) \quad (2)$$

where $\phi(\cdot)$ is the all-MiniLM-L6-v2 sentence encoder (Reimers and Gurevych 2019) loaded from the local HuggingFace cache (no network access required), and the max is over all sentences in the chunk. Atom a_i is classified ENTAILMENT if $\max_j s(a_i, \mathcal{C}_j) \geq \tau_e$ (threshold $\tau_e = 0.42$), NEUTRAL if $\geq \tau_n = 0.25$, and CONTRADICTION otherwise. Numeric claims containing values not present in any retrieved chunk are down-graded from ENTAILMENT to NEUTRAL.

$$\text{SummaC-ZS}(c, \mathcal{C}) = \frac{|\{a_i : \text{label}(a_i) = E\}|}{m} \quad (3)$$

Tier 2: SCR (Semantic Coverage Rate). Whole-claim vs. best-chunk cosine similarity (a coarser, complementary signal): $\text{SCR}(c, \mathcal{C}) = \max_j \cos(\phi(c), \phi(\mathcal{C}_j))$.

Tier 3: KFP (Key Fact Presence). Fraction of numeric values and technical proper nouns from the claim found verbatim in the top-5 retrieved chunks (exact lexical recall for quantities and named entities, where semantics cannot substitute).

The combined score is:

$$\text{CFS} = 0.50 \times \text{SummaC-ZS} + 0.30 \times \text{SCR} + 0.20 \times \text{KFP} \quad (4)$$

4.9 System Architecture

Scholar is built as a fully local system. The backend is a FastAPI application that stores parsed PDFs as JSON files on disk (`metadata.json`, `pages.json`, `chunks.json`). The frontend is a Next.js single-page application with an embedded pdf.js viewer that highlights cited pages in real time. All inference, both text generation and figure/table visual question answering, is performed locally via Ollama using a single natively multimodal model (qwen3.5:9b or gemma4:12b); no cloud API is used anywhere in the pipeline. No user data ever leaves the local machine. All experiments in this paper were run on a consumer laptop (Apple Silicon, 18 GB unified memory) under Python 3.12, PyTorch, and Ollama, with all-MiniLM-L6-v2 (Reimers and Gurevych 2019) loaded from a local Hugging Face cache for the faithfulness pipeline; no GPU or cloud compute was used for any reported result.

5 Evaluation

5.1 Retrieval Benchmark

We evaluate retrieval quality on a manually annotated benchmark of 14 cases spanning three landmark NLP papers: *Attention Is All You Need* (1706.03762 (Vaswani et al. 2017)), *Retrieval-Augmented Generation* (2005.11401 (Lewis et al.

Table 1: Visual grounding retrieval results (18 cases). All cases correctly route to the target figure or table.

Reasoning Type	N	Retrieval R@5
Visual understanding	6	1.000
Data extraction	5	1.000
Comparison	4	1.000
Location	3	1.000
Overall	18	1.000

2020)), and *LLaMA* (2302.13971 (Touvron et al. 2023)). Each case specifies a user-style question, a set of expected relevant chunk IDs, and the expected source pages. Query types include main idea, architecture, training details, result tables, human evaluation, safety analysis, and page-hint queries. We report Recall@ K ($K \in \{1, 3, 5\}$) and Mean Reciprocal Rank (MRR).

5.2 Faithfulness Benchmark

For the faithfulness evaluation, we construct **51 oracle-claim cases** manually annotated across three landmark NLP papers: *Attention Is All You Need* (Vaswani et al. 2017) (15 cases), *RAG* (Lewis et al. 2020) (12 cases), and *LLaMA* (Touvron et al. 2023) (24 cases). Each case contains a natural-language query, a ground-truth claim, a manually verified supporting chunk ID, and a claim type label. Claim types include: `result_number` (13), `training_detail` (8), `technical_claim` (11), `architecture_detail` (10), `formula` (1), `conceptual_claim` (5), `human_eval` (2), and `environmental_claim` (1). All annotations were produced manually by the authors.

5.3 Visual Grounding Benchmark

We construct an 18-case pilot benchmark spanning four reasoning types: `visual_understanding` (6), `data_extraction` (5), `comparison` (4), and `location` (3). Each case specifies a figure- or table-targeted question and the expected figure/table label(s) that should be retrieved. We report two signals: *retrieval grounding* — whether the correct figure or table is ranked first, isolating the visual-cue retrieval and figure-number matching mechanism (Section 4.5) from the vision model’s reading accuracy — and a lightweight *answer-quality* proxy (the fraction of non-trivial content words from the question itself that reappear in the generated answer), which does depend on the vision model. Under gemma4:12b, retrieval grounding is 18/18 and every case is answered directly by the vision model with mean answer score 0.731 (no case fell back to the caption-only degraded path). Table 1 reports retrieval results by reasoning type. We deliberately keep this benchmark small and frame it as a pilot study: it validates that the routing mechanism works and gives a first read on answer quality, not that visual question answering is solved.

Table 2: Multi-document locality results. `locality_arxiv` (10 cases) is the primary metric; `detail` cases confirm anchor-paper retrieval is unaffected by the larger multi-paper pool.

Subset	<i>N</i>	R@1	R@5	MRR
All	18	0.222	0.611	0.364
Locality (all)	15	0.067	0.533	0.237
Locality (arXiv)	10	0.100	0.800	0.356
Detail (anchor)	3	1.000	1.000	1.000

Table 3: Multi-document locality: ScholAR vs. a random-guess floor and an oracle upper bound (10 `locality_arxiv` cases).

System	R@1	R@5	MRR
Random floor (no signal)	0.125	0.625	0.340
ScholAR (flat retrieval)	0.100	0.800	0.356
Oracle (correct paper isolated)	0.700	0.900	0.783

5.4 Multi-Document Locality Benchmark

We construct an 18-case multi-document benchmark in the style of M3SciQA (Li et al. 2024): 15 *locality* cases, each requiring the system to identify which cited reference (not the anchor paper) contains the answer, and 3 *detail* cases answerable from the anchor paper alone as a sanity check. Of the 15 locality cases, 10 reference works with a resolvable arXiv identifier (`locality_arxiv`, our primary metric); the remaining 5 reference pre-arXiv works or datasets (e.g., WMT, Dropout, Wikipedia) that cannot currently be downloaded and always score zero. Table 2 reports Recall@*K* and MRR for each subset.

The correct cited paper is usually recoverable within the top 5 results (80%) but rarely ranks first (10%): with dozens of papers’ chunks merged into a single flat retrieval pool, a niche secondary paper must win on lexical match alone against the anchor paper’s typically larger, more topically central chunk set. Section 6 discusses this finding relative to M3SciQA’s comparable observation that even large foundation models struggle at this exact task.

To give this number honest context rather than leaving it unexplained, we compute two reference bounds on the 10 `locality_arxiv` cases (average 8.0 candidate secondary papers per case): a *random floor* — the expected recall of picking a candidate paper uniformly at random with no retrieval signal at all — and an *oracle* upper bound, where retrieval is restricted to only the anchor paper plus the single correct secondary paper (i.e., the paper-selection problem is assumed solved, isolating chunk-level retrieval precision *within* the correct paper). Table 3 reports all three.

Two things stand out. First, ScholAR’s actual R@1 (0.100) is *below* the random-guess floor (0.125) — at the single-guess precision level, flat retrieval across a large multi-paper pool is not just imperfect, it is slightly worse than chance, almost certainly because the anchor paper’s larger, more BM25-friendly chunk set systematically wins the top rank regard-

Table 4: Visual grounding under two local models (18 cases each). Answer score is a keyword-overlap heuristic, not an LLM-judge score.

Model	Retrieval R@5	Answer Score	Fallback
<code>gemma4:12b</code>	1.000	0.731	0/18
<code>qwen3.5:9b</code>	1.000	0.812	0/18

less of which secondary paper is actually most relevant. At R@5 (0.800 vs. a 0.625 floor), the same flat retrieval clearly does carry a real signal above chance once enough slots are available for it to surface. Second, the oracle bound (R@1 0.700, MRR 0.783) shows that most of the gap between ScholAR and a perfect system is a *paper-selection* problem, not a chunk-level retrieval problem: once the correct paper is isolated, within-paper retrieval is nearly as strong as the single-document benchmark in Table 5. This directly motivates the per-paper-aware retrieval and RLM directions discussed next, rather than further tuning of chunk-level ranking.

5.5 Local Model Comparison

Retrieval and faithfulness scoring in ScholAR are model-independent (BM25 and a fixed local sentence encoder, with no LLM call in either path), so the only benchmark where the choice of local Ollama model is actually measurable is visual grounding, where the vision-capable model both reads the figure and writes the answer. We run the 18-case visual grounding benchmark (Section 5.3) under two natively multimodal local models, `gemma4:12b` and `qwen3.5:9b`, holding everything else fixed. Table 4 reports retrieval grounding and a keyword-overlap answer-quality proxy for each. Both models retrieve the correct figure in all 18 cases and never fall back to the caption-only degraded path, but the smaller `qwen3.5:9b` (9B parameters) scores higher on answer quality than the larger `gemma4:12b` (12B parameters) on this benchmark. We do not read this as a general claim that smaller models are better — the benchmark is small and the scoring proxy is a lexical heuristic, not an LLM-judge — but it does show that going fully local does not require the largest available model, and that model choice is worth benchmarking empirically per task rather than assumed from parameter count alone.

5.6 Does Reading the Image Help? A Caption-Only Ablation

Rather than implement a full ColPali-style (Faysse et al. 2024) visual embedding baseline, which would require new retrieval infrastructure this benchmark is not built to compare against, we run a cheaper, more direct ablation: send the identical prompt to the same local model *without* the figure image attached, and score the answer with the same heuristic. Under `gemma4:12b`, caption-only scores **0.752** versus **0.731** for full vision — removing the image does not lower this particular score.

We do not read this as evidence that visual reading is unnecessary. Many benchmark questions ask about a value also

Table 5: Retrieval results on 14 manually annotated cases. PH = page hints. Best values in **bold**.

System	R@1	R@3	R@5	MRR
Keyword Overlap	0.571	0.786	0.929	0.687
BM25 Only	0.714	0.857	0.929	0.788
BM25+Rerank (No PH)	0.714	0.857	0.929	0.788
BM25+Rerank (PH)	0.714	0.857	0.929	0.788
Dense-Only	0.786	1.000	1.000	0.881

restated in the paper’s prose near the figure, so a text-only answer can be correct without consulting the image; and our answer-quality proxy rewards lexical overlap with the *question* rather than grounded correctness, so a model that reports fine-grained visual details absent from the question wording (e.g., a specific symbol connecting two boxes in a diagram, observed qualitatively in Section 4.5) is not credited for them. Properly separating these effects requires benchmark cases whose answer is genuinely absent from the paper’s prose, plus a correctness-based judge rather than a lexical-overlap proxy; we flag this as necessary follow-up work rather than overstate what the current ablation supports.

5.7 Baseline Systems

For retrieval, we compare five configurations: (1) **Keyword Overlap**: Jaccard-style term overlap with page-hint boost; (2) **BM25-Only**: standard BM25 with page-hint boost; (3) **BM25+Rerank (No PH)**: BM25 without page hints; (4) **BM25+Rerank (PH)**: full system with page hints (default); (5) **Dense-Only**: single-pass `all-MiniLM-L6-v2` cosine similarity, no BM25, no rerank, no page hints — the naive-RAG point in the lexical-to-dense taxonomy (Thakur et al. 2021). For faithfulness, we compare: (6) **BM25-primary**: the ScholAR production retriever; (7) **Hybrid BM25+Dense+RRF**: dense retrieval fused via Reciprocal Rank Fusion (Cormack, Clarke, and Buettcher 2009).

5.8 Retrieval Results

Table 5 shows the retrieval results. BM25-primary outperforms keyword overlap on R@1, R@3, and MRR, and ties it on R@5. Page hints do not change the aggregate MRR on this benchmark but improve retrieval for specific page-reference queries. Dense-Only unexpectedly beats every lexical baseline (R@1 0.786, R@3/R@5 1.000, MRR 0.881), contrary to BEIR’s finding that zero-shot dense retrieval trails BM25 (Thakur et al. 2021), likely an artifact of this benchmark’s small size and paraphrased queries. At faithfulness scale (51 cases, Table 6) hybrid, not dense-only, wins, so we do not read this as grounds to change the production retriever.

5.9 Faithfulness Results

Table 6 reports SummaC-ZS faithfulness results on the 51-case benchmark. The hybrid BM25+Dense+RRF system achieves a Combined CFS of **0.829**, improving over BM25-primary (0.809) by 2.0 percentage points. More significantly, hybrid retrieval improves SCHR@5 from 0.824 to **0.922**, a gain of 9.8 percentage points, confirming that dense semantic

Table 6: SummaC-ZS faithfulness results on 51 oracle-claim cases. T1 = SummaC-ZS entailment fraction; SCR = whole-claim cosine similarity; KFP = Key Fact Presence; SCHR@ K = Supporting Chunk Hit Rate.

Metric	BM25-primary	Hybrid+RRF
SummaC-ZS / T1 (entailment)	0.941	0.961
SCR / T2 (cosine similarity)	0.484	0.509
KFP / T3 (key fact recall)	0.966	0.979
Combined CFS	0.809	0.829
SCHR@1	0.373	0.392
SCHR@3	0.725	0.686
SCHR@5	0.824	0.922
Faithful (≥ 0.55 CFS)	48 / 51	49 / 51

Table 7: Combined CFS at each retrieval depth ($N = 51$ cases). Both improve monotonically; hybrid leads at every depth except a near-tie at Top-3.

Chunks	N	BM25	Hybrid
Top-1	51	0.658	0.669
Top-2	51	0.757	0.776
Top-3	51	0.799	0.796
Top-4	51	0.791	0.817
Top-5	51	0.809	0.829

retrieval recovers chunks that BM25 lexical matching misses. BM25-primary classifies 48 of 51 cases (94.1%) and Hybrid **49 of 51** (96.1%) as FAITHFUL under the ≥ 0.55 strict entailment threshold. Both systems’ SummaC-ZS Tier 1 entailment (0.941 / 0.961) and KFP (0.966 / 0.979) scores are high, with SCR (whole-claim cosine) remaining the primary differentiating factor between systems.

5.10 Ablation: CFS vs. Retrieval Depth

Table 7 shows how Combined CFS grows monotonically as more retrieved chunks are included. This confirms that evidence for complex scientific claims is distributed across multiple pages and that returning Top-5 chunks is necessary to achieve full faithfulness coverage. The hybrid system leads at most depths, with the largest gap at Top-1 (BM25 0.658 vs. Hybrid 0.669); the two are within 0.3 points of each other at Top-3, where BM25 briefly edges ahead, before hybrid pulls back into the lead at Top-4 and Top-5.

5.11 Ablation: CFS by Claim Type

Table 8 breaks down Combined CFS by claim type under BM25-primary. Environmental and human evaluation claims score highest (0.918, 0.882), as their supporting text is verbally rich and semantically dense. Training-detail claims (0.668) are penalised by multi-hop evidence spread across sections; the two chronic BM25 failure cases (`attn_training_steps`: 0.242 UN-FAITHFUL; `llama_training_tokens`: 0.356 PARTIAL) both involve numerical claims spread across mul-

Table 8: CFS breakdown by claim type, BM25-primary system, $N=51$.

Claim Type	N	BM25 CFS	Hybrid CFS
Environmental claim	1	0.918	0.918
Human evaluation	2	0.882	0.882
Result number	13	0.811	0.860
Conceptual claim	5	0.847	0.854
Technical claim	11	0.846	0.856
Architecture detail	10	0.830	0.845
Formula	1	0.876	0.876
Training detail	8	0.668	0.674

tuple non-contiguous page ranges. A third BM25 failure, `rag_trivia_score` (0.235 UNFAITHFUL, a `result_number` claim), is retrieval-induced rather than a distributed-evidence case: BM25 misses the supporting chunk entirely in its Top-5 for this query, while Hybrid still recovers it (0.802 FAITHFUL) via the dense component — see the discussion of the retrieval-precision/compound-term-matching tradeoff below.

6 Discussion

Hybrid retrieval improves semantic recall. BM25-primary achieves Recall@5 of 0.929 and MRR of 0.788, confirming that lexical matching over page-preserving chunks is a reliable foundation for scientific document QA. Our SummaC-ZS evaluation further reveals that hybrid dense retrieval adds meaningful value: `SCHR@5` improves from 0.824 to 0.922 (+9.8 pp) and Combined CFS rises from 0.809 to 0.829 (+2.0 pp). These gains are driven by the dense component surfacing semantically relevant chunks even when exact BM25 query terms are absent.

SummaC-ZS detects paraphrased entailment that ROUGE cannot. Our Tier 1 SummaC-ZS score (0.961 for both systems) is significantly higher than a purely lexical approach would yield for the same cases. The key distinction is that SummaC-ZS uses sentence-level cosine similarity (via `all-MiniLM-L6-v2`) which captures semantic equivalence across paraphrased formulations, a property that ROUGE-family metrics fail to exhibit (Zha et al. 2023). The SCR tier (whole-claim cosine, 0.484 BM25 vs. 0.509 Hybrid) is the primary differentiating signal between the two retrieval systems.

Training-detail claims are the main failure mode. The two non-FAITHFUL cases (`attn_training_steps`, `llama_training_tokens`) are both training-detail claims where numerical evidence is distributed across non-contiguous page ranges. This validates structured, math-aware extraction as the highest-priority future work for the single-document pipeline.

Retrieval depth matters. The rank ablation shows monotonic CFS improvement from Top-1 (BM25: 0.658, Hybrid: 0.669) to Top-5 (0.809 / 0.829), confirming the ScholAR default of $K=5$ is well-motivated.

Tokenizer fixes trade off single-document precision against compound-term matching; hybrid retrieval ab-

sorbs the cost better than BM25 alone. We fixed a tokenizer bug uncovered during multi-document testing: token boundaries only start on a letter, so a fused compound proper noun (e.g. a figure or paper name written as one capitalized word) and its space-separated form reduce to different token sets, breaking exact-name retrieval. The fix is necessary for Sections 4.5 and 4.6, but it perturbs BM25’s length normalisation globally: it cost BM25-only Recall@5 seven points ($1.00 \rightarrow 0.929$) and introduced one new faithfulness retrieval miss (`rag_trivia_score`). Hybrid BM25+Dense+RRF was materially more robust — its numbers barely moved, and it still recovers the case BM25 now misses — showing hybrid retrieval’s real robustness to exactly this kind of perturbation.

Cross-paper localization is the multi-document pipeline’s weak point. The multi-document locality benchmark (Section 5.4) shows a wide gap between Recall@5 (0.80) and Recall@1 (0.10): the correct secondary paper is usually retrieved but rarely ranks first. We attribute this to the flat retrieval design — all loaded papers’ chunks share one BM25 pool with no per-paper normalisation, so a niche cited paper competes directly against the anchor paper’s larger, more central chunk set. This mirrors M3SciQA (Li et al. 2024), where even large foundation models underperform human experts at the same cross-paper localization task. We view per-paper-aware retrieval/reranking and the Recursive Language Model pattern (Zhang, Kraska, and Khattab 2025) as the most promising directions for closing this gap.

Baseline scope. Section 2.2 explains why OpenScholar, PaperQA2, and SciRAG are discussed qualitatively rather than run head-to-head: none targets a single consumer laptop, so a fair comparison needs a resource-matched setup (e.g., PaperQA2 on a local Ollama backend), which we leave to future work rather than an untested, crippled substitute or a cloud-API run that would contradict our design constraint.

7 Future Work

The training-detail failure mode identified in Section 5.2 (numerical evidence spread across non-contiguous pages) motivates structured extraction as the highest-priority next step: we plan to integrate Nougat (Blecher et al. 2023) for LaTeX-preserving extraction and evaluate it against ColPali (Faysse et al. 2024)-style visual page retrieval as an alternative to our current caption-and-crop figure grounding. A math-aware tokenizer that indexes LaTeX tokens (e.g., `\sigma`, `\frac`) alongside natural language terms remains future work for the BM25 retrieval layer, though rendered math display in the answer UI is already implemented via KaTeX. We also plan to evaluate the Recursive Language Model inference pattern (Zhang, Kraska, and Khattab 2025) as an alternative to flat retrieval for multi-document citation-chasing, where our current locality benchmark shows the clearest room for improvement.

8 Conclusion

We presented ScholAR, a fully local retrieval-augmented system for grounded scientific document comprehension, extending single-document QA to visual grounding and multi-

document citation-chasing via page-preserving segmentation and indirect citation grounding. Across four benchmarks: BM25-primary retrieval reaches Recall@5 of 0.929 (14 cases); our NLI-CFS faithfulness pipeline shows hybrid BM25+Dense+RRF raising Combined CFS to 0.829 and SCHR@5 to 0.922 (51 cases), with training-detail claims hardest; visual grounding routes correctly in all 18 pilot cases; and multi-document locality reaches 80% Recall@5 but only 10% at top-1 — a flat-retrieval limitation shared with comparable systems, and the clearest open problem for future work.

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